

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1096

Hofman 100 kV
December 15, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1096. The Interconnection Customer proposes a 59.9 MW natural gas combined cycle facility interconnecting at the Hofman 100 kV substation in the Indiana area. The request was received on November 02, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 421 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,851,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of MONTPELIER ELECTRIC GENERATING STA 230/1 kV Autotransformer, the POWERLINE ELKHORN - WEBSTERVILLE 100 KV loads to 159.6% of its 134 MVA rating. Project DFAX: 6.1%.

For the loss of MONTPELIER ELECTRIC GENERATING STA 230-1 kV Autotransformer, the POWERLINE ELKHORN - THOUSAND SPRINGS 100 KV line loads to 154.6% of its 127 MVA rating. Project DFAX: 6.1%.

COLUMBIA HILLS SUBSTATION - CGUANAS 8103 100 KV line: 143.0% loading (89 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STA 230/1kV Auto; DFAX = 6.3%.

For the loss of MONTPELIER ELECTRIC GENERATING STA 230-1 kV TX, the ELDEAN - MONASHEE 100 KV loads to 142.1% of its 136 MVA rating. Project DFAX: 5.9%.

For the loss of E METHUEN 500/230kV TX, the STAUNTON - PIQUA POWER PLANT 230 KV line loads to 133.6% of its 517 MVA rating. Project DFAX: 5.1%.

COSO 7 - MONTPELIER ELECTRIC GENERATING STATION 100 KV Line: 129.9% loading (156 MVA) following the outage of the MUSKOGEE - MONTPELIER ELECTRIC GENERATING STATION 500 KV; DFAX = 13.3%.

ELDEAN - CARBONDALE NW 100 KV Line: 128.5% loading (165 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STATION 230-1 kV Auto; DFAX = 5.9%.

The CARLEY ROAD - PIQUA POWER PLANT 230 KV is overloaded to 128.0% of its 540 MVA emergency rating for the loss of the EAST METHUEN 500-230 kV Auto. The project contributes 5.1% DFAX (3.1 MW) to this overload.

For the loss of E METHUEN SUBSTATION 500 kV/230 kV Autotransformer, the STAUNTON - BARAGA VG 230 KV loads to 125.4% of its 532 MVA rating. Project DFAX: 5.1%.

For the loss of MONTPELIER ELECTRIC GENERATING STA 230/1kV TX, the SHEEPSKIN - CAGUANAS 8103 100 KV Line loads to 124.7% of its 109 MVA rating. Project DFAX: 6.3%.

For the loss of MONTPELIER ELECTRIC GENERATING STATION SUBSTATION 230 kV/1 kV Transformer Bank, the RICHARD LAKE - CARBONDALE NW 100 KV loads to 122.6% of its 192 MVA rating. Project DFAX: 5.9%.

For the loss of SWAN VALLEY - TREGO 500 KV, the TREGO - WEST MELROSE 230 KV L loads to 122.1% of its 623 MVA rating. Project DFAX: 6.4%.

For the loss of MUSKOGEE - SWAN VLY 500 KV, the MUSKOGEE 500-230 kV Auto loads to 121.2% of its 1,410 MVA rating. Project DFAX: 16.0%.

The SAVONA - RICHARD LAKE 100 KV L is overloaded to 118.2% of its 219 MVA emergency rating for the loss of the MONTPELIER ELECTRIC GENERATING STA 230/1kV TX. The project contributes 5.9% DFAX (3.6 MW) to this overload.

MUSKOGEE - TIMBER SWITCH 230 KV Line: 117.7% loading (731 MVA) following the outage of the SWAN VALLEY - TREGO 500 KV line; DFAX = 6.5%.

The WEST MELROSE - TIMBER SWITCH 230 KV is overloaded to 116.4% of its 770 MVA emergency rating for the loss of the SWAN VALLEY - TREGO 500 KV. The project contributes 6.4% DFAX (3.8 MW) to this overload.

For the loss of MONTPELIER ELECTRIC GENERATING STA 230 kV/1 kV XFMR, the MONASHEE - WEST RIVERSIDE 100 KV line loads to 116.4% of its 145 MVA rating. Project DFAX: 5.9%.

SAVONA - BUTLER MUNICIPAL POWER PLANT 100 KV L: 115.8% loading (237 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STATION 230 kV/1 kV Auto; DFAX = 5.9%.

W ASHEVILLE - WEBSTERVILLE 100 KV line: 115.6% loading (125 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STA SUBSTATION 500/230kV TX; DFAX = 5.1%.

For the loss of MONTPELIER ELECTRIC GENERATING STA 230/1kV TX, the BUTLER MUNICIPAL POWER PLANT - HANSEL 100 KV L loads to 114.1% of its 253 MVA rating. Project DFAX: 5.9%.

The HANSEL - THOUSAND SPRINGS 100 KV is overloaded to 113.0% of its 263 MVA emergency rating for the loss of the MONTPELIER ELECTRIC GENERATING STATION 230/1 kV TX. The project contributes 5.9% DFAX (3.6 MW) to this overload.

For the loss of MONTPELIER ELECTRIC GENERATING STATION 230-1 kV Auto, the DUTTON PUMP - LEYENDECKER SUBSTATION 100 KV L loads to 112.1% of its 144 MVA rating. Project DFAX: 5.5%.

For the loss of WABASH 500 kV/230 kV TX, the MONTPELIER ELECTRIC GENERATING STA - SIGNEICS 100 KV loads to 108.2% of its 142 MVA rating. Project DFAX: 38.3%.

For the loss of LONETREE - E METHUEN 500 KV, the LONETREE 500/230 kV XFMR loads to 107.1% of its 1,165 MVA rating. Project DFAX: 6.6%.

The HENRY COUNTY 500/230kV TX is overloaded to 106.0% of its 1,903 MVA emergency rating for the loss of the LONETREE - E METHUEN 500 KV. The project contributes 12.7% DFAX (7.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
POWERLINE ELKHORN - WEBSTERVILLE 100 KV	Rebuild line with higher-capacity conductor	\$16,855,000	14% (\$2,405,000)
POWERLINE ELKHORN - THOUSAND SPRINGS 100 KV	Rebuild line with higher-capacity conductor	\$1,404,000	14% (\$200,000)
CAGUANAS 8103 - COLUMBIA HILLS 100 KV	Rebuild line with higher-capacity conductor	\$4,463,000	35% (\$1,563,000)
ELDEAN - MONASHEE SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$4,095,000	10% (\$389,000)
STAUNTON - PIQUA POWER PLANT 230 KV line	Rebuild line with higher-capacity conductor	\$13,687,000	0% (\$71,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COSO 7 - MONTPELIER ELECTRIC GENERATING STATION 100 KV Line	Rebuild line with higher-capacity conductor	\$11,204,000	26% (\$2,863,000)
CARBONDALE NW - ELDEAN 100 KV	Rebuild line with higher-capacity conductor	\$1,950,000	10% (\$185,000)
CARLEY ROAD - PIQUA POWER PLANT 230 KV	Rebuild line with higher-capacity conductor	\$41,436,000	0% (\$216,000)
STANTON - BARAGA VLG 230 KV	Rebuild line with higher-capacity conductor	\$28,619,000	0% (\$149,000)
CAGUANAS 8103 - SHEEPSKIN 100 KV	Rebuild line with higher-capacity conductor	\$5,243,000	35% (\$1,836,000)
RICHARD LAKE - CARBONDALE NW 100 KV	Rebuild line with higher-capacity conductor	\$3,099,000	10% (\$294,000)
TREGO - WEST MELROSE 230 KV L	Rebuild line with higher-capacity conductor	\$20,924,000	1% (\$179,000)
MUSKOGEE 500-230 kV Auto	Replace transformer with higher-rated unit	\$31,553,000	1% (\$377,000)
SAVONA - RICHARD LAKE 100 KV L	Rebuild line with higher-capacity conductor	\$9,698,000	10% (\$921,000)
MUSKOGEE - TIMBER SWITCH 230 KV Line	Rebuild line with higher-capacity conductor	\$25,000,000	1% (\$215,000)
W MELROSE - TIMBER SWITCH 230 KV Line	Rebuild line with higher-capacity conductor	\$165,299,000	1% (\$1,429,000)
MONASHEE - W RIVERSIDE 100 KV	Rebuild line with higher-capacity conductor	\$4,538,000	10% (\$431,000)
SAVONA - BUTLER MUNICIPAL POWER PLANT 100 KV L	Rebuild line with higher-capacity conductor	\$1,015,000	10% (\$96,000)
W ASHEVILLE - WEBSTERVILLE 100 KV line	Rebuild line with higher-capacity conductor	\$37,480,000	10% (\$3,781,000)
BUTLER MUNICIPAL POWER PLANT - HANSEL 100 KV L	Rebuild line with higher-capacity conductor	\$4,252,000	10% (\$404,000)
HANSEL - THOUSAND SPRINGS 100 KV	Rebuild line with higher-capacity conductor	\$8,924,000	10% (\$847,000)
DUTTON PUMP - LEYENDECKER 100 KV	Rebuild line with higher-capacity conductor	\$9,433,000	12% (\$1,151,000)
MONTPELIER ELECTRIC GENERATING STA - SIGNEICS 100 KV	Reconductor line	\$744,000	82% (\$607,000)
LONETREE 500/230 kV Auto	Replace transformer with higher-rated unit	\$17,275,000	1% (\$136,000)
HENRY COUNTY 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$23,807,000	0% (\$106,000)
	Total allocated to project		\$20,851,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.